



7240 - Gas cooling in the molecular CGM of the Spiderweb: in-situ galaxy growth at $z=2$

Cycle: 4, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Bjorn Emonts (PI)	Associated Universities, Inc.
Dr. Pierre Guillard (CoI) (ESA Member) (CoPI) (Contact)	Institut d'Astrophysique de Paris
Dr. Matthew D. Lehnert (CoI) (ESA Member) (CoPI)	Centre de Recherche Astrophysique de Lyon
Dr. Patrick Michael Ogle (CoI)	Space Telescope Science Institute
Dr. Carlos Albert De Breuck (CoI) (ESA Member)	European Southern Observatory - Germany
Dr. Philip N. Appleton (CoI)	California Institute of Technology
Dr. Ilsang Yoon (CoI)	National Radio Astronomical Observatory
Prof. Nina Hatch (CoI) (ESA Member)	University of Nottingham
Prof. Guillaume Pineau des Forets (CoI) (ESA Member)	Institut d'Astrophysique Spatiale
Dr. Helmut Dannerbauer (CoI) (ESA Member)	Instituto de Astrofisica de Canarias
Dr. Maria Montserrat Villar Martin (CoI) (ESA Member)	Centro de Astrobiologia - CAB
Dr. Lars E. Kristensen (CoI) (ESA Member)	University of Copenhagen, Niels Bohr Institute
Jorge Villa-Velez (CoI) (ESA Member)	Ecole Normale Supérieure
Dr. Francois Boulanger (CoI) (ESA Member)	Ecole Normale Supérieure
Sophie N Lebowitz (CoI)	University of Arizona
Dr. George K. Miley (CoI) (ESA Member)	Universiteit Leiden
Dr. Benjamin Godard (CoI) (ESA Member)	Observatoire de Paris
Dr. Zheng Cai (CoI)	Tsinghua University
Mingyu Li (CoI)	Tsinghua University
Prof. Aaron S. Evans (CoI)	The University of Virginia
Dr. Jose Manuel Perez Martinez (CoI) (ESA Member)	Instituto de Astrofisica de Canarias

<i>Name</i>	<i>Institution</i>
Niusha Ahvazi (CoI)	The University of Virginia
Dr. Thomas Lai (CoI)	California Institute of Technology

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	SW-ON	MIRI Medium Resolution Spectroscopy	(1) SW-GALAXY
	2	SW-ON	MIRI Medium Resolution Spectroscopy	(1) SW-GALAXY
	3	SW-OFF	MIRI Medium Resolution Spectroscopy	(2) SW-OFF
	4	SW-OFF	MIRI Medium Resolution Spectroscopy	(2) SW-OFF

ABSTRACT

The evolution of galaxies at Cosmic Noon is linked to complex physical processes that take place across the circum-galactic medium (CGM), such as gas outflows, accretion, and mixing. However, a direct link between the CGM and the stellar growth of galaxies critically depends on identifying cooling processes that trigger the formation of molecular clouds - and ultimately stars - on large scales. Deep HST and ALMA imaging showed that the massive Spiderweb Galaxy at $z=2.2$ is rapidly forming stars in-situ from cold molecular gas across the CGM. The goals of this JWST proposal are to understand how the CGM can cool down to form stars, and on what scale in the CGM this process happens. JWST/MIRI is uniquely suited to study this through line observations of warm H₂, which unambiguously trace the dissipation of kinetic energy from the CGM. We will use this to test the hypothesis that H₂ is the dominant coolant in the CGM. Using the MRS, we will image the distribution and kinematics of the pure rotational H₂ 0-0 lines to study the H₂ excitation, energy budget, and cooling rates across the CGM. This will be compared with a suite of molecular lines imaged with ALMA, including new ALMA Band 1 data of CO(1-0), which traces the coldest H₂ gas. Commensally, we will obtain deep MIRI wideband imaging, and complement this with existing deep (19 orbits) HST imaging, to derive masses, rates and time-scales of the in-situ star formation. By studying H₂ as the main channel for widespread cooling, and comparing cooling and star-formation rates across the CGM, we will gain a comprehensive understanding of the critical role that the multi-phase CGM plays the early evolution of massive galaxies.

OBSERVING DESCRIPTION

We will use the MIRI MRS for a single pointing in subbands B and C to obtain a deep exposure that covers the H₂ 0-0 rotational lines across the circum-galactic medium (CGM) of the Spiderweb Galaxy in Channels 3 and 4. This setup will also capture the main H₂ 1-0 and 2-1 ro-vibrational lines, together with other interesting lines and PAH features, in Channels 1 and 2. We require separate background calibration exposures with the MIRI MRS, which we will center such that we will obtain simultaneous deep MIRI imaging of our target with the F560W and F1000W filters, which

JWST Proposal 7240 (Created: Thursday, March 26, 2026, 4:00:11PM Eastern Standard Time) - Overview

will complement existing deep HST imaging of the spatially resolved stellar populations across the CGM of the Spiderweb.

Proposal 7240 - Targets - Gas cooling in the molecular CGM of the Spiderweb: in-situ galaxy growth at $z=2$

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	SW-GALAXY	RA: 11 40 48.2922 (175.2012175d) Dec: -26 29 8.63 (-26.48573d) Equinox: J2000		
	Comments: This object was generated by the targetselector and retrieved from the NED database. Category=Galaxy Description=[Radio galaxies] Extended=YES				
Fixed Targets	(2)	SW-OFF	RA: 11 40 51.2800 (175.2136667d) Dec: -26 28 40.00 (-26.47778d) Equinox: J2000		
	Comments: This object was generated by the targetselector and retrieved from the NED database. Category=Galaxy Description=[Radio galaxies] Extended=YES				

Proposal 7240 - Observation 1 - Gas cooling in the molecular CGM of the Spiderweb: in-situ galaxy growth at z=2

Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	SLOWR1	12	10	1	Dither 1	4	40	12327.199	222551.28
	1	MEDIUM(B)	MRSLONG		SLOWR1	30	9	1	Dither 1	4	36	26565.591	222551.24
	1	MEDIUM(B)	MRSSHORT		SLOWR1	30	9	1	Dither 1	4	36	26565.591	222551.34
Special Requirements	Sequence Observations 1, 3, Non-interruptible												

Proposal 7240 - Observation 2 - Gas cooling in the molecular CGM of the Spiderweb: in-situ galaxy growth at z=2

Observation	Proposal 7240, Observation 2: SW-ON				Thu Mar 26 21:00:11 GMT 2026
	Diagnostic Status: Warning				
	Observing Template: MIRI Medium Resolution Spectroscopy				
	Background Observations:[SW-OFF (Obs 4)]				
	Comments: REDUCED EXPOSURE TIME IMAGING: We reduced the exposure time of the MIRI imaging during the SW-ON visit, because there is a 12.8 mag star in the outer part of the field, and the images taken during SW-ON will not be used for science. We wish to avoid strong straylight and saturating the detector during the ON observation to avoid latents in the OFF observations, given that the non-interruptable OFF observations will perform long-exposure science imaging. Note that for the OFF observations the bright star was positioned outside the F.o.V. of the MIRI imager, hence maximum integration time is requested for imaging during the OFF visit. DATA RATES: Our visits comply with the recommended maximum data volumes, but the data rates (in Mb/s) are over the currently recommended limit for FASTR1. The FASTR1 mode would imply Nggroups >= 100, so we use the SLOWR1 readout mode to comply with data rate limits. SCHEDULING WINDOWS AND PA range: both observing windows (May-July and Dec-Feb) are acceptable, but Dec-Feb falls in the Meteoroid Avoidance Zone, so we optimized the position of the MRS field of view for May-July (PA range between 80 and 122 degrees), to capture the relevant regions. In case of scheduling during the secondary window (Dec-Feb, PA=290-330), we would need to (1) slightly re-tune the pointing position of SW-ON (by up to a few arcsec at most), and (2) re-position the background MRS calibration field to capture the target in the MIRI imager for executing the proposed commensal science. WORKBOOK IDs: Integration times for both ON and OFF are based on the H2 0-0 S(5) line in MRS Long - Medium (B). This integration time will also allow us to detect the lines of interest in MRS Short and Long (C). The corresponding ETC workbooks are: 222551.24 (SNR=9 for H2 0-0 S(5) in the CGM with r=0.8 arcsec aperture); 222551.30 (SNR=23 for H2 0-0 S(5) in the galaxy with r=0.8 arcsec aperture); 222551.34 (SNR=11 for Pa-alpha in the CGM with r=0.5 arcsec aperture); 222551.33 (SNR=55 for Pa-alpha in the galaxy with r=0.5 arcsec aperture). The integration time for commensal imaging is set by the exposure time for MRS background calibration (and shortened to mitigate the effect of a bright star for the ON exposures). ETC workbook 222551.28 for imaging is included for completion only.				
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	SW-GALAXY	RA: 11 40 48.2922 (175.2012175d) Dec: -26 29 8.63 (-26.48573d) Equinox: J2000 Comments: This object was generated by the targetselector and retrieved from the NED database. Category=Galaxy Description=[Radio galaxies] Extended=YES		
Acquisition	#	Target			
	1	NONE			
Template	AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	F560W	Channel 3	YES	FULL	Allow Auto Reorder
Dithers	#	Dither Type	Optimized For	Direction	
	1	4-Point	EXTENDED SOURCE	NEGATIVE	

Proposal 7240 - Observation 2 - Gas cooling in the molecular CGM of the Spiderweb: in-situ galaxy growth at z=2

Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/E xp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F1000W	SLOWR1	12	10	1	Dither 1	4	40	12327.199	222551
	1	LONG(C)	MRSLONG		SLOWR1	30	9	1	Dither 1	4	36	26565.591	222551
	1	LONG(C)	MRSSHORT		SLOWR1	30	9	1	Dither 1	4	36	26565.591	222551
Special Requirements	Sequence Observations 2, 4, Non-interruptible												

Proposal 7240 - Observation 3 - Gas cooling in the molecular CGM of the Spiderweb: in-situ galaxy growth at z=2

Observation	Proposal 7240, Observation 3: SW-OFF				Thu Mar 26 21:00:11 GMT 2026
	Diagnostic Status: Warning				
	Observing Template: MIRI Medium Resolution Spectroscopy				
	Background Observation For: [SW-ON (Obs 1)]				
	Comments: DATA RATES: Our visits comply with the recommended maximum data volumes, but the data rates (in Mb/s) are over the currently recommended limit for FASTR1. The FASTR1 mode would imply Ngroups >= 100, so we use the SLOWR1 readout mode to comply with data rate limits. DITHERING: We request EXTENDED 4-pt dithering also for the OFF field in order to (1) optimize the simultaneous MIRI science imaging, and (2) to optimize the MIRI MRS background subtraction for reaching a high contrast on the faint large-scale emission. SCHEDULING WINDOWS AND PA range: both observing windows (May-July and Dec-Feb) are acceptable, but Dec-Feb falls in the Meteoroid Avoidance Zone, so we optimized the position of the MRS field of view for May-July (PA range between 80 and 122 degrees), to capture the relevant regions. In case of scheduling during the secondary window (Dec-Feb, PA=290-330), we would need to (1) slightly re-tune the pointing position of SW-ON (by up to a few arcsec at most), and (2) re-position the background MRS calibration field to capture the target in the MIRI imager for executing the proposed commensal science. WORKBOOK IDs: Integration times for both ON and OFF are based on the H2 0-0 S(5) line in MRS Long - Medium (B). This integration time will also allow us to detect the lines of interest in MRS Short and Long (C). The corresponding ETC workbooks are: 222551.24 (SNR=9 for H2 0-0 S(5) in the CGM with r=0.8 arcsec aperture); 222551.30 (SNR=23 for H2 0-0 S(5) in the galaxy with r=0.8 arcsec aperture); 222551.34 (SNR=11 for Pa-alpha in the CGM with r=0.5 arcsec aperture); 222551.33 (SNR=55 for Pa-alpha in the galaxy with r=0.5 arcsec aperture). The integration time for commensal imaging is set by the exposure time for MRS background calibration (and shortened to mitigate the effect of a bright star for the ON exposures). ETC workbook 222551.28 for imaging is included for completion only.				
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.				
Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(2)	SW-OFF	RA: 11 40 51.2800 (175.2136667d) Dec: -26 28 40.00 (-26.47778d) Equinox: J2000 Comments: This object was generated by the targetselector and retrieved from the NED database. Category=Galaxy Description=[Radio galaxies] Extended=YES		
Acquisition	#	Target			
	1	NONE			
Template	AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	F560W	Channel 3	YES	FULL	Allow Auto Reorder
Dithers	#	Dither Type	Optimized For	Direction	
	1	4-Point	EXTENDED SOURCE	NEGATIVE	

Proposal 7240 - Observation 3 - Gas cooling in the molecular CGM of the Spiderweb: in-situ galaxy growth at z=2

Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	SLOWR1	12	21	1	Dither 1	4	84	25992.233	222551.28
	1	MEDIUM(B)	MRSLONG		SLOWR1	30	9	1	Dither 1	4	36	26565.591	222551.24
	1	MEDIUM(B)	MRSSHORT		SLOWR1	30	9	1	Dither 1	4	36	26565.591	222551.34
Special Requirements	Sequence Observations 1, 3, Non-interruptible												

Proposal 7240 - Observation 4 - Gas cooling in the molecular CGM of the Spiderweb: in-situ galaxy growth at z=2

Observation	Proposal 7240, Observation 4: SW-OFF					Thu Mar 26 21:00:11 GMT 2026
	Diagnostic Status: Warning					
	Observing Template: MIRI Medium Resolution Spectroscopy					
	Background Observation For: [SW-ON (Obs 2)]					
	Comments: DATA RATES: Our visits comply with the recommended maximum data volumes, but the data rates (in Mb/s) are over the currently recommended limit for FASTR1. The FASTR1 mode would imply Ngroups >= 100, so we use the SLOWR1 readout mode to comply with data rate limits. DITHERING: We request EXTENDED 4-pt dithering also for the OFF field in order to (1) optimize the simultaneous MIRI science imaging, and (2) to optimize the MIRI MRS background subtraction for reaching a high contrast on the faint large-scale emission. SCHEDULING WINDOWS AND PA range: both observing windows (May-July and Dec-Feb) are acceptable, but Dec-Feb falls in the Meteoroid Avoidance Zone, so we optimized the position of the MRS field of view for May-July (PA range between 80 and 122 degrees), to capture the relevant regions. In case of scheduling during the secondary window (Dec-Feb, PA=290-330), we would need to (1) slightly re-tune the pointing position of SW-ON (by up to a few arcsec at most), and (2) re-position the background MRS calibration field to capture the target in the MIRI imager for executing the proposed commensal science. WORKBOOK IDs: Integration times for both ON and OFF are based on the H2 0-0 S(5) line in MRS Long - Medium (B). This integration time will also allow us to detect the lines of interest in MRS Short and Long (C). The corresponding ETC workbooks are: 222551.24 (SNR=9 for H2 0-0 S(5) in the CGM with r=0.8 arcsec aperture); 222551.30 (SNR=23 for H2 0-0 S(5) in the galaxy with r=0.8 arcsec aperture); 222551.34 (SNR=11 for Pa-alpha in the CGM with r=0.5 arcsec aperture); 222551.33 (SNR=55 for Pa-alpha in the galaxy with r=0.5 arcsec aperture). The integration time for commensal imaging is set by the exposure time for MRS background calibration (and shortened to mitigate the effect of a bright star for the ON exposures). ETC workbook 222551.28 for imaging is included for completion only.					
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.					
Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous	
	(2)	SW-OFF	RA: 11 40 51.2800 (175.2136667d) Dec: -26 28 40.00 (-26.47778d) Equinox: J2000 Comments: This object was generated by the targetselector and retrieved from the NED database. Category=Galaxy Description=[Radio galaxies] Extended=YES			
Acquisition	#	Target				
	1	NONE				
Template	AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction	
	F560W	Channel 3	YES	FULL	Allow Auto Reorder	
Dithers	#	Dither Type	Optimized For	Direction		
	1	4-Point	EXTENDED SOURCE	NEGATIVE		

Proposal 7240 - Observation 4 - Gas cooling in the molecular CGM of the Spiderweb: in-situ galaxy growth at z=2

Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/E xp	Exposures/Dit h	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F1000W	SLOWR1	12	21	1	Dither 1	4	84	25992.233	222551
	1	LONG(C)	MRSLONG		SLOWR1	30	9	1	Dither 1	4	36	26565.591	222551
	1	LONG(C)	MRSSHORT		SLOWR1	30	9	1	Dither 1	4	36	26565.591	222551
Special Requirements	Sequence Observations 2, 4, Non-interruptible												